

Manish Dwibedy

EXPERIENCE

Petco – Remote *Senior Engineer*

May 2024 – Present

- **Architected and optimized high-performance, low-latency APIs** for a SaaS platform using **Python** and **Flask**, achieving significant response time improvements via **Redis** caching.
- **Engineered scalable, cost-effective distributed systems** by designing and implementing a serverless, event-driven architecture using **AWS Lambda, SQS, S3, and API Gateway**.
- **Automated critical operational workflows** (e.g., dynamic pricing and SKU generation), directly enhancing the efficiency and speed of physical store deployments.
- **Ensured SaaS platform reliability** and rapid incident response through the implementation of comprehensive monitoring and alerting systems (**Sumo Logic** and **PagerDuty**).
- **Acted as a technical leader** in cross-functional teams, translating product vision and UI/UX requirements into actionable technical specifications.
- **Technologies:** Python, Flask, FastAPI, React, PostgreSQL, Docker, AWS (Lambda, SQS, S3, API Gateway), Sumo Logic, PagerDuty.

Social Protocol – Remote *Lead Engineer*

July 2022 – April 2024

- **Led full-stack development and architecture** for two major initiatives, reporting directly to the CEO on technical strategy.
- **Developed an AI/LLM Aggregator Platform**, integrating models (OpenAI, Bard) for a unified AI service interface.
- **Engineered a custom Retrieval-Augmented Generation (RAG) solution** using in-house hosting to optimize smaller language models.
- **Explored and integrated advanced text and image LLMs** to enhance platform generative AI capabilities.
- **Built a vertical-specific platform** (like Zillow) for the religious sector, managing complex data and UX.
- **Designed the complete technical architecture**, including backend, frontend, database, and hosting infrastructure.
- **Developed robust RESTful APIs and backend tasks**, utilizing **MongoDB** for efficient data storage.
- **Created dynamic and responsive user interfaces** using the **React** framework.
- **Managed the technical roadmap and progress**, presenting strategic recommendations to the CEO.
- **Technologies:** Python, Flask, Node.js, React, MongoDB, Docker, Vercel, Google Maps.

The Trade Desk – Remote *Senior Software Engineer*

May 2022 - June 2022

- **Led development for the new 'Data Access' platform**, serving internal and external clients with high-performance APIs.
- **Implemented diverse service interfaces** including **RESTful**, **GraphQL**, and **gRPC** to ensure flexible data consumption across the organization.
- **Leveraged modern microservices technologies** for deployment and orchestration, ensuring scalability and reliability.
- **Technologies Used:** Python, FastAPI, Postgres, Docker, and Kubernetes.

Tatras Data – Module Lead & *Senior Software Engineer*

Sept 2020 – May 2022

- Led backend API development and collaborated with data scientists to **feed ETL/data pipelines**, returning model outputs for PathFactory's **content recommendation engine**.
- Engineered **asynchronous operations using AWS Lambda** to handle long-running Facebook API

requests for Good United's Donation Campaign contributions.

- Contributed to backend API and database operations utilizing **Amazon RDS (PostgreSQL)**, ensuring data integrity and performance.
- Provided **technical mentorship** to junior engineers and actively participated in candidate interviewing and evaluation.
- Collaborated weekly with the onsite product development team to ensure feature alignment and project velocity.
- Technologies: Python, FastAPI, Celery, Redis, MongoDB, SQL, AWS, and Docker.

Scry Analytics Inc – San Jose, California

May 2018 – Apr 2020

- **Led development for Cisco's IoT and AI Detection Platforms**, including creating APIs for AI communication, deploying solutions on Cisco hardware, and developing the platform UI.
- **Integrated with high-profile Cisco clients** ("San Francisco 49ers," "Southern California Edison"), delivering customized platform solutions.
- **Contributed to ABB's Predictive Maintenance AI Pipeline**, using **Python/PySpark** and collaborating with Data Scientists to implement anomaly detection for factory machinery failure prediction.
- **Engineered the data pipeline and logic** for **ABB's** AI solution, optimizing data flow and model input/output.
- **Developed the 'Collatio' Document Processing Platform**, handling full-stack tasks (backend and frontend) to extract data from various documents.
- **Executed two high-impact Proof of Concepts (POCs):**
 - **Computer Vision solution Employee Detection** for a potential Japanese banking client.
 - **Anti-Money Laundering (AML) Detection** in collaboration with the data science team for **Wells Fargo**.
- **Delivered multiple platform modules** using a hybrid backend stack (**Python** and **Node.js**) alongside an **Angular** dashboard.
- **Managed cross-timezone collaboration** with teams in Gurugram and Shanghai to ensure continuous project alignment.
- **Technologies:** Python, Node.js, Angular 8, MongoDB, AWS, Azure Cloud, and Docker.

Social Spaces (Internship) – Los Angeles, California

Jan 2018 - Apr 2018

- **Developed an Event Recommendation System** focused on enhancing user engagement and content discovery.
- **Assisted in the development and implementation of the platform's blockchain component** on the **Ethereum (ETH)** platform.

Very Green Ventures (Internship) – Los Angeles, California

Oct 2017 - Dec 2017

- **Integrated and implemented payment processing** for the client-side web application using the **Stripe** API.
- **Developed an Admin Dashboard** to visualize application activity and display key performance analytics.
- **Built and deployed a cross-platform mobile application** using **React Native**.

Single Focus Inc. (Internship) – Los Angeles, California

Mar 2017 - Oct 2017

- **Developed, tested, and successfully launched the mobile application on the Apple App Store.**
- **Implemented core application services, including the authentication service and the Map-based view.**

Premo LLC(Internship) – Los Angeles, California

Jun 2016 - Aug 2016

- **Served as a Full Stack Developer**, specializing in **iOS application development**.

- **Developed and tested core application modules**, including **user management**, the **payment platform**, and **messaging functionality**.
- Utilized **Swift** for native iOS development and **Node.js** for handling backend payment processing logic.

Big Data Day LA (Internship) – Los Angeles, California

Apr 2016 - Jul 2016

- **Developed and launched the official iOS event application**, providing attendees with features like **schedule**, **talks**, and **speaker information**.
- **Engineered the event registration system**, successfully facilitating attendance for over **1,000 attendees**.

University of Southern California – Los Angeles, California

Jan 2015 - May 2017

- **Assisted Professor Marco Papa** for **CSCI 571: Web Technologies**, supporting over 200 students.
- **Provided academic support** by holding office hours, grading assignments, and helping with exam preparation.
- **Gained expertise in the course stack**, including **PHP**, **Angular**, **iOS App development**, and **Google Cloud**.

Tata Consultancy Services (TCS) – Gurugram, India

Sept 2012 - Jun 2015

- **Led the development of "HOBBS Connected Devices Management,"** a core product for managing network devices within telecom networks.
- **Engineered the backend API and developed the accompanying product dashboard** for device management and visualization.
- **Developed, deployed, and rigorously tested multiple product modules** prior to a successful launch for Vodafone Australia.
- Contributed to technical talent development by **mentoring junior engineers** and participating in **recruitment activities**.
- Technologies: Java/J2EE, MySQL, PERL, and UI development (JSP, HTML, CSS, JavaScript).

SKILLS

Backend - Python (Flask, FastAPI, strong proficiency), Node.JS

Data - SQL (Postgres), NoSQL (MongoDB, DynamoDB), Redis and Kafka

Frontend - HTML/CSS3/JS, jQuery, Bootstrap, Angular, React and Flutter

Infrastructure & DevOps: **Terraform**, **Docker**, Kubernetes, **CircleCI** (CI/CD), **Serverless** architecture, **AWS** (Lambda, SQS, EC2, DynamoDB, API Gateway, CloudFront, RDS, EKS, VPC, CloudWatch, S3, IAM)

Additional Experience: Solidity Smart Contracts, AI/ML Fundamentals, Large Language Models (LLMs)

CERTIFICATIONS & TRAININGS

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|---|----------|
| • Java Standard Edition 6 Programmer Certified Professional by Oracle | Mar 2015 |
| • GraphQL Professional Certification by Apollo GraphQL | Dec 2024 |
| • Sumo Logic: Search Mastery, Metrics Mastery, and Fundamentals | Dec 2024 |

EDUCATION

University of Southern California

Aug 2015- May 2017

Master of Science, Computer Science, Computer Science; GPA: 3.61/4.0

Guru Gobind Singh Indraprastha University

Aug 2008-May 2012

Bachelor of Technology, Computer Science, GPA - 3.7/4.0

PERSONAL INFORMATION

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ACHIEVEMENT

ADPList - Software Development Mentor

Jan 2023 - Present

- Mentoring and guiding college undergraduates and junior software engineers build their careers in the most efficient manner.

QualiMed (Solo Hackathon)

Sep 2025

- QualiMed is an AI-driven tool that uses an LLM to automate the creation of comprehensive test cases from various software requirements (text, PDF, DOCX), significantly streamlining QA and boosting test coverage.
- Its USP is a privacy-focused, offline, and highly customizable AI, allowing users to select between cloud APIs or local models and fine-tune behavior to specific project needs.
- Prototype: <https://qualimed-805085509940.us-west1.run.app>
- Video Demo: <https://youtu.be/uxCcmrQLMHY>
- Source Code: <https://github.com/manishdwibedy/QualiMed>

StorySpark AI (Solo Hackathon)

Aug 2025

- StorySpark AI is an AI-Powered Storybook Video Creator that transforms text prompts into narrated stories, utilizing Gemini 2.5 Flash Image for visuals and ElevenLabs for high-quality audio.
- Key Features: Used Gemini for character consistency and image editing. Integrated ElevenLabs for human-like narration and included customization for story frame management.
- Prototype: <https://storyspark-ai-805085509940.us-west1.run.app>
- Video Demo: <https://youtu.be/uxCcmrQLMHY>
- Source Code available on request

eInsure

April 2020

- Developed a product to provide better visibility in the InsurTech domain.
- **Solo developer**, pivoted to work in the supply chain domain instead.
- Won the **5th position finish in World Blockchain Hackathon (Online)**.

Novagen

Feb 2018

- Developed a Flask web based application to discover new prescription drugs.
- Team of 4, Source code Available at <https://github.com/michael13162/novogen>
- Won the **Best Data Science in Health Field in TreeHacks 2018 at Stanford, CA**.